

Mid-term Exam (Graph Neural Networks - Fall 2026)

Full Name:

Student ID:

Note: Students should write in English for evaluation.

9. (Node2vec) Consider an undirected graph G given in the following figure. A random walk has just moved from node A to node B , with $p = 2$ and $q = 0.5$.

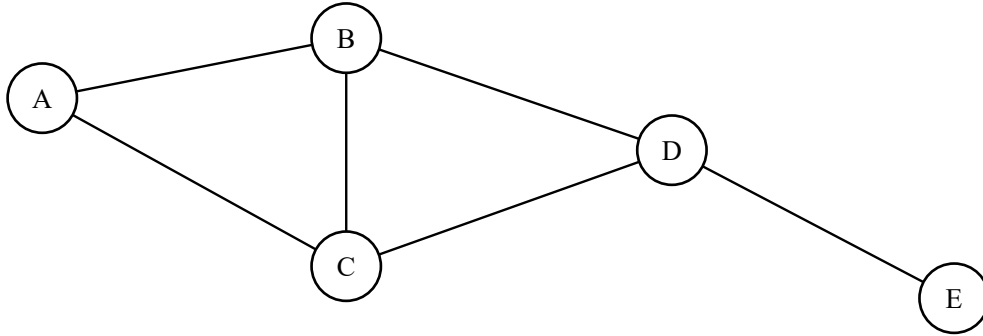


Figure 1. Graph G

The unnormalized transition probability from B to a next node x is $\pi_{Bx} = \alpha_{pq}(A, x)$, where

$$\alpha_{pq}(A, x) = 1/p \text{ if } d(A, x) = 0; \quad 1 \text{ if } d(A, x) = 1; \quad 1/q \text{ if } d(A, x) = 2$$

and $d(A, x)$ is the shortest path distance between node A and node x .

- Explain BFS and DFS, and explain which one a small q makes the walk behave like.
- List all possible next nodes from node B and calculate the unnormalized transition probabilities π_{Bx} .
- Calculate the normalized transition probabilities.

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10. (LINE) Consider an undirected graph G with node embeddings

$$u_A = 1, u_B = 0.5, u_C = -0.5, u_D = -1$$

and context embeddings

$$u'_A = 0.5, u'_B = 1, u'_C = -1, u'_D = -0.5.$$

The second-order proximity is defined as

$$p_2(j|x) = \exp(u'_j u_x) / \sum_k \exp(u'_k u_x).$$

Given $e^{0.25} = 1.28$, $e^{0.5} = 1.65$, $e^{-0.25} = 0.78$, $e^{-0.5} = 0.61$.

- a) What is second-order proximity, and how does it differ from first-order proximity?
- b) Calculate $p_2(A|B)$, $p_2(B|B)$, $p_2(C|B)$ and $p_2(D|B)$.

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11. (GCN) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 1 \\ -1 \\ 2 \\ 0 \\ 3 \end{bmatrix}$$

Assume that the hidden layer of a GCN model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \sigma(\tilde{A} \cdot H^{(k-1)}),$$

where $H^{(k)}$ denotes the output at layer k , $\tilde{A}$ is the normalized matrix ($\tilde{A} = D^{-1}A$), and σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

- a) Explain how information is propagated in a GCN layer.
- b) Calculate $\tilde{A}$.
- c) Calculate the output of the GCN model at layer $k = 1$.

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12. (GraphSAGE) Consider an undirected graph G of five nodes A, B, C, D and E given in the following figure. Each node has initial features that are the numbers standing next to it (i.e. the initial feature of node 'A' is $h_A^{(0)} = 0.4$). According to the GraphSAGE model with an AGGREGATE that is a MEAN function, the feature of a node i at layer k can be updated as:

$$h_{N(i)}^{(k)} = \text{AGGREGATE}(\{ h_u^{(k-1)}, \forall u \in N(i) \})$$

$$h_i^{(k)} = \text{ReLU}(h_i^{(k-1)} \parallel h_{N(i)}^{(k)})$$

where $\parallel$ is a concatenation, $\text{ReLU}(x) = \max(0, x)$, and $N(i)$ is the set of neighbour nodes of node i .

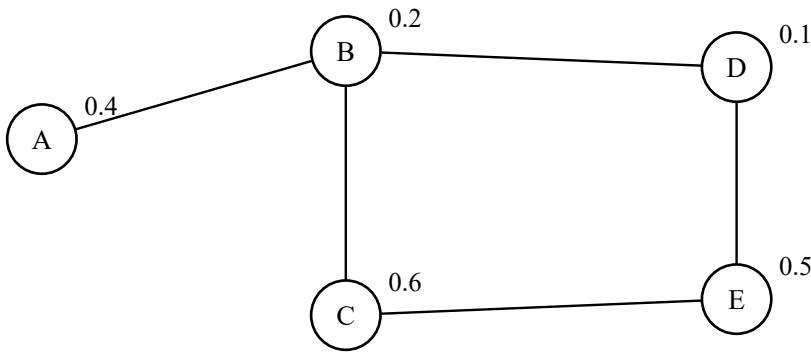


Figure 2. Graph G

- What is the main idea behind the GraphSAGE model, and how does it differ from traditional GCNs?
- Calculate the feature of each node at $k = 1$.
- Calculate a graph-level embedding h_G by using a 'Mean' global pooling when $k = 1$.

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13. (ClusterGCN) Consider an undirected graph G of six nodes A, B, C, D, E and F given in the following figure. The graph G contains two clusters G_1 and G_2 . Each node has initial features that are the numbers standing next to it. According to the ClusterGCN model, the feature of a node i at layer k can be updated as:

$$h_{N(i)}^{(k)} = \text{MEAN}(\{ h_u^{(k-1)}, \forall u \in N(i), G_u = G_i \})$$

$$h_i^{(k)} = \text{ReLU}(h_i^{(k-1)} \parallel h_{N(i)}^{(k)})$$

where $\parallel$ is a concatenation and $\text{ReLU}(x) = \max(0, x)$.

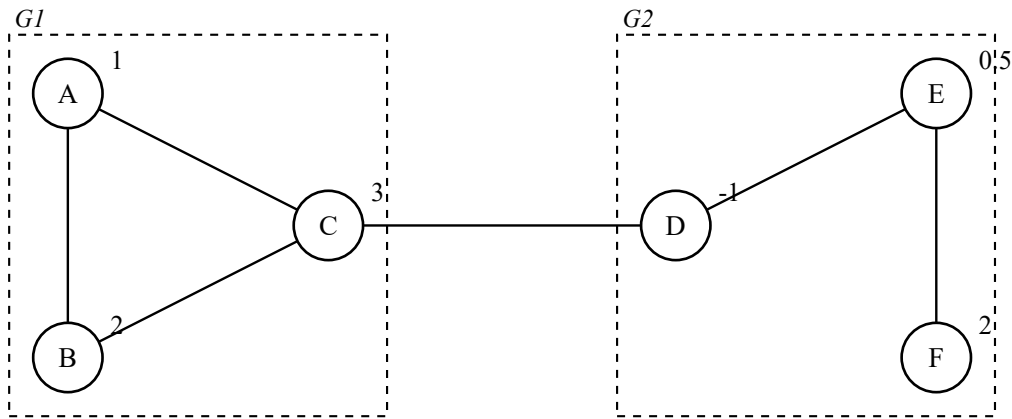


Figure 3. Graph G with two clusters

- What is mini-batch training, and why does ClusterGCN need it?
- Calculate the output representations of all nodes at layer $k = 1$.

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14. (GraphSAINT) Consider an undirected graph G of six nodes A, B, C, D, E and F given in the following figure. The graph G has a sampled subgraph G_1 . Each node has initial features that are the numbers standing next to it. According to the GraphSAINT model, the feature of a node i at layer k can be updated as:

$$h_{N(i)}^{(k)} = \text{MEAN}(\{ h_u^{(k-1)}, \forall u \in N(i), G_u = G_i \})$$

$$h_i^{(k)} = \text{ReLU}(h_i^{(k-1)} \parallel h_{N(i)}^{(k)})$$

where $\parallel$ is a concatenation and $\text{ReLU}(x) = \max(0, x)$.

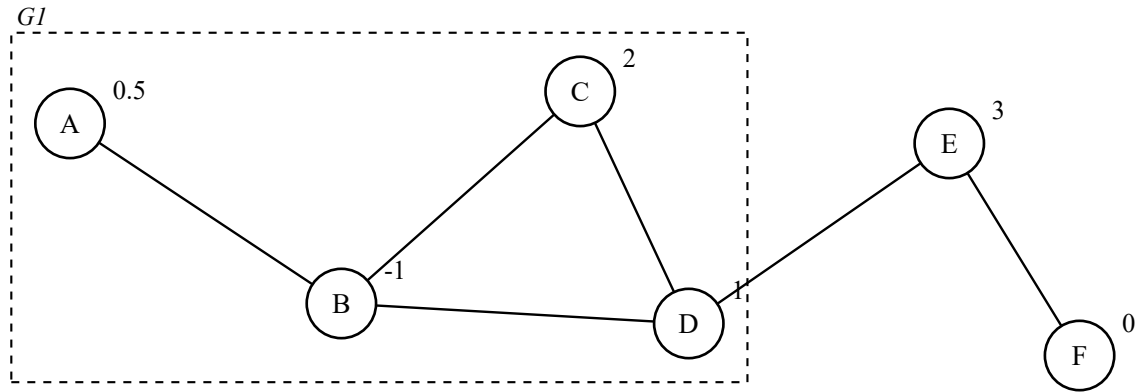


Figure 4. Graph G with a sampled subgraph G_1

- a) How does GraphSAINT's sampling strategy differ from node-wise or layer-wise sampling methods like GraphSAGE and FastGCN?
- b) Calculate the output representations of nodes A, B, C and D at layer $k = 1$.

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15. (JK Network) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} -1 \\ 0 \\ 1 \\ -1 \\ 2 \end{bmatrix}$$

Assume that the output of a JK network model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \max(\sigma(\tilde{A} \cdot H^{(0)}), \sigma(\tilde{A} \cdot H^{(1)}), \dots, \sigma(\tilde{A} \cdot H^{(k-1)}))$$

where $H^{(k)}$ denotes the output at layer k , $\tilde{A}$ is the normalized matrix ($\tilde{A} = D^{-1}A$), and σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

- a) Explain how Jumping Knowledge Networks help mitigate the over-smoothing problem in deep GCNs.
- b) Calculate $\tilde{A}$.
- c) Calculate the output representations at layer $k = 2$.

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16. (GCNII) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 3 \\ -1 \\ 1 \\ 2 \\ -3 \end{bmatrix}$$

Assume that the output of a GCNII model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \sigma[((1 - \beta) I_n) \cdot ((1 - \alpha) \tilde{A} \cdot H^{(k-1)} + \alpha H^{(0)})]$$

where $H^{(k)}$ denotes the output at layer k , $\tilde{A}$ is the normalized matrix ($\tilde{A} = D^{-1}A$), I_n is the identity matrix, $\alpha = \beta = 0.5$, and σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

- a) What are the two major components introduced in GCNII to overcome the depth limitation problem in GCNs?
- b) Calculate $\tilde{A}$.
- c) Calculate the output representations at layer $k = 1$.

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17. (DeepGCN) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 2 \\ -2 \\ 1 \\ 0 \end{bmatrix}$$

Assume that the output of a DeepGCN model with a residual connection of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \sigma(\tilde{A} \cdot H^{(k-1)}) + H^{(k-1)}$$

where $\tilde{A} = D^{-1}A$ and σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

- a) Explain why a residual connection allows a GCN to be made deeper.
- b) Calculate the output representations at layer $k = 1$ and $k = 2$.
- c) Compare $H^{(1)}$ with the output of a plain GCN without the residual connection.

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1. (Relational Classification) Consider an undirected graph G of five nodes given in the following figure. Nodes 1 and 5 are labelled ($P(Y_1) = 1, P(Y_5) = 0$) and the others are unlabelled and initialized to 0.5. The relational classifier updates every unlabelled node as:

$$P(Y_v = c) = (\sum_{(v,u) \in E} A_{v,u} P(Y_u = c)) / (\sum_{(v,u) \in E} A_{v,u})$$

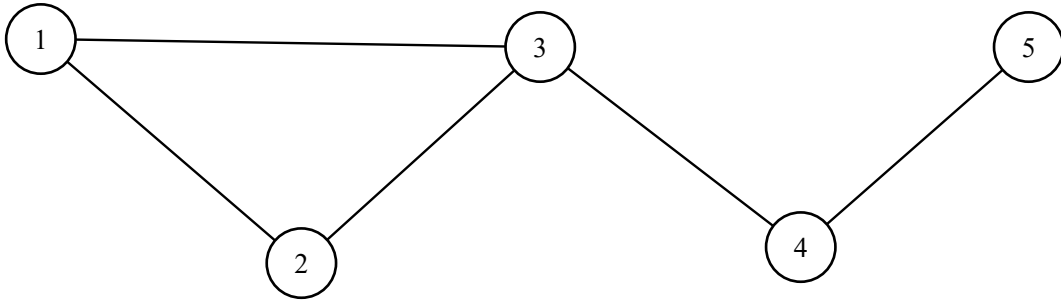


Figure 5. Graph G

- Explain the key assumption behind relational classification, and state one limitation of it.
- Calculate $P(Y_v)$ for every unlabelled node after the 1st iteration.
- Calculate $P(Y_v)$ after the 2nd iteration. Update nodes in the order 2, 3, 4 using the values from the previous iteration only.

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2. (FastGCN) Consider the undirected graph G with adjacency matrix A and node feature vector $h^{(0)}$ below. FastGCN replaces the full neighbourhood sum of a GCN layer by a layer-wise Monte-Carlo estimate:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$h^{(0)} = \begin{bmatrix} 1 \\ 2 \\ -1 \\ 0 \\ 3 \end{bmatrix}$$

$$h^{(l+1)}(v) = \sigma \left(\frac{1}{t_l} \sum_{j=1}^{t_l} \tilde{A}(v, u_j) h^{(l)}(u_j) \right)$$

where $\tilde{A} = D^{-1}A$, the t_l nodes u_j are drawn once per layer from the importance distribution $q(u) \propto \|\tilde{A}(:, u)\|^2$, and $\text{ReLU}(x) = \max(0, x)$.

- a) How does layer-wise sampling in FastGCN differ from the node-wise sampling of GraphSAGE, and what problem does it solve?
- b) Calculate the importance distribution $q(u)$ over the five nodes.
- c) Suppose the sampled set for the layer is $\{B, C\}$ ($t_l = 2$). Calculate the estimate of $h^{(1)}(A)$.

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3. (ASGCN) Consider the same graph G as in question 11, with $\tilde{A} = D^{-1}A$ and the node feature vector $h^{(0)}$ below. ASGCN rewrites a GCN layer as an expectation:

$$h^{(0)} = \begin{bmatrix} 1 \\ 2 \\ -1 \\ 0 \\ 3 \end{bmatrix}$$

$$h^{(l+1)}(v_i) = \sigma(W^{(l)} N(v_i) E_{p(u_j | v_i)}[h^{(l)}(u_j)])$$

where $N(v_i) = \sum_j \tilde{A}_{i,j}$ is the normalization factor and $p(u_j | v_i) = \tilde{A}_{i,j} / N(v_i)$ is the probability of sampling node u_j given the parent node v_i . Take $W^{(l)} = 1$ and $\text{ReLU}(x) = \max(0, x)$.

- a) ASGCN is called 'adaptive' sampling. What does the proposal distribution depend on in ASGCN that it does not depend on in FastGCN?
- b) Calculate $N(v_i)$ and $p(u_j | v_i)$ for the parent node $v_i = B$.
- c) Calculate $E_{p(u_j | B)}[h^{(0)}(u_j)]$ and the resulting $h^{(1)}(B)$.

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4. (APPNP) Given a graph with an adjacency matrix A and a prediction matrix H produced by an MLP as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$H = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

APPNP (Approximate Personalized Propagation of Neural Predictions) propagates the predictions with a teleport probability α :

$$Z^{(0)} = H, \quad Z^{(k+1)} = (1 - \alpha) \tilde{A} Z^{(k)} + \alpha H$$

where $\tilde{A} = D^{-1}A$ and $\alpha = 0.5$. No nonlinearity is applied between propagation steps.

- a) Explain the teleport mechanism of Personalized PageRank and why it prevents over-smoothing even with many propagation steps.
- b) Calculate $\tilde{A}$ and $Z^{(1)}$.
- c) Calculate $Z^{(2)}$.

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5. (GPR-GNN) Given a graph with an adjacency matrix A and hidden features $H^{(0)}$ produced by a neural network as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 2 \\ 0 \\ 1 \\ -1 \end{bmatrix}$$

GPR-GNN (Generalized PageRank GNN) separates feature extraction from propagation:

$$H^{(k)} = \tilde{A} H^{(k-1)}, \quad Z = \sum_{k=0}^K \gamma_k H^{(k)}$$

where $\tilde{A} = D^{-1}A$, the weights γ_k are learned, and here $K = 2$ with $\gamma_0 = 0.5$, $\gamma_1 = 0.3$, $\gamma_2 = -0.2$.

- a) What can GPR-GNN do that APPNP cannot, and why does allowing negative γ_k matter for heterophilic graphs?
- b) Calculate $H^{(1)}$ and $H^{(2)}$.
- c) Calculate Z .

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6. (MixHop) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 1 \\ 2 \\ 0 \\ -1 \end{bmatrix}$$

A MixHop layer concatenates several powers of the adjacency matrix:

$$H^{(i+1)} = \mathbb{I}_{j \in P} \sigma(\hat{A}^j H^{(i)} W_j^{(i)})$$

where $\mathbb{I}$ is a column-wise concatenation, $P = \{0, 1, 2\}$, $\hat{A} = D^{-1}A$, $W_j^{(i)} = 1$ for every j , and $\text{ReLU}(x) = \max(0, x)$.

- a) State Theorem 1 of MixHop about the two-hop Delta Operator, and explain why a vanilla GCN cannot represent it even with infinitely many layers.
- b) Calculate $\hat{A}^0 H^{(0)}$, $\hat{A}^1 H^{(0)}$ and $\hat{A}^2 H^{(0)}$.
- c) Calculate $H^{(1)}$, and calculate the Delta Operator $\sigma(\hat{A}^0 H^{(0)}) - \sigma(\hat{A}^2 H^{(0)})$.

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7. (PairNorm) After one GCN layer the node features of a four-node graph are

$$X = \begin{bmatrix} 3 & 1 \\ 1 & -1 \\ -1 & 2 \\ 1 & 2 \end{bmatrix}$$

PairNorm normalizes them in two steps, a centering step and a scaling step:

$$x_i^c = x_i - (1/n) \sum_{i=1}^n x_i \quad (\text{Center})$$

$$x_i = s \cdot x_i^c / \text{sqrt}((1/n) \sum_{i=1}^n \|x_i^c\|_2^2) \quad (\text{Scale})$$

with $s = 1$ and $n = 4$.

- a) Explain node-wise and feature-wise over-smoothing, and explain which quantity PairNorm keeps constant across layers.
- b) Calculate the centered features x_i^c .
- c) Calculate the scaling denominator and the final x_i .

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8. (DropEdge) Consider the complete graph K_4 on nodes A, B, C and D with the adjacency matrix A and initial node feature matrix $H^{(0)}$ below.

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 2 \\ 4 \\ -2 \\ 0 \end{bmatrix}$$

DropEdge keeps each undirected edge independently with probability $1 - p$ at every training epoch, and the layer is then computed on the surviving graph:

$$H^{(1)} = \sigma((D^{(t)})^{-1} A^{(t)} H^{(0)})$$

where $A^{(t)}$ and $D^{(t)}$ are rebuilt from the surviving edges only, $p = 1/3$, and $\text{ReLU}(x) = \max(0, x)$.

- a) Which two problems does DropEdge target, and how does it differ from Dropout?
- b) Calculate the expected number of surviving edges, and calculate $H^{(1)}$ on the full graph.
- c) Suppose the edges (A, D) and (B, C) are dropped in this epoch. Recalculate $H^{(1)}$.

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18. (GAT) Consider an undirected graph G of four nodes A, B, C and D. Each node has an initial feature next to it ($h_A^{(0)} = 0.3$, $h_B^{(0)} = 0.1$, $h_C^{(0)} = 0.2$, $h_D^{(0)} = 0.2$). The weight matrix is $W = [0.5]$.

$$h_i^{(k)} = \sigma(\sum \alpha_{im} W h_m), \quad m \in N(i)$$

$$\alpha_{im} = e_{im} / \sum e_{ik}, \quad e_{im} = \sigma(\text{MEAN}(W h_i, W h_m))$$

where σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$. Node A is connected to B, C and D.

- a) What problem in traditional GCNs does GAT aim to solve?
- b) Calculate the attention coefficients e_{AB} , e_{AC} and e_{AD} .
- c) Calculate the feature of node A at $k = 1$.

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19. (GATv2) Compare the attention score e_{ij} of GAT and GATv2.

$$GAT: \quad e_{ij} = \text{LeakyReLU}(a(W h_i, W h_j))$$

$$GATv2: \quad e_{ij} = a \text{ LeakyReLU}(W[h_i, h_j])$$

$\text{vec } h_i = [1, 2]^T$, $\text{vec } h_j = [1, -1]^T$, $W = I_2$, $a = [1, 1, 1, 1]$, and $\text{LeakyReLU}(x) = x$ if $x > 0$, $0.02x$ if $x < 0$.

- a) What is the key limitation of GAT that GATv2 aims to fix?
- b) Calculate e_{ij} for both and compare.

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20. (HAN) Consider a heterogeneous academic graph with three node types: Author (A), Paper (P) and Venue (V). Initial features are $h_{P1} = 0.3$, $h_{P2} = 0.1$, $h_{P3} = 0.2$, $h_{A1} = 0.1$, $h_{A2} = 0.1$, $h_{V1} = 0.2$, and $W = [0.5]$. Edges:

$P1 - A1$, $P2 - A1$, $P2 - A2$, $P3 - A2$, $P1 - V1$, $P3 - V1$.

$$h_i^{(k)} = \sigma(\sum \alpha_{im} W h_m), \quad m \in N(i)$$

$$\alpha_{im} = e_{im} / \sum e_{ik}, \quad e_{im} = \sigma(\text{MEAN}(W h_i, W h_m))$$

The superscript Φ means the meta-path. σ is a ReLU function.

- a) What is a heterogeneous graph?
- b) List all meta-paths PAP and PVP, and calculate the attention coefficients.
- c) Calculate the feature of node $P1$ at $k = 1$.

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21. (Homophily Metrics) Consider the undirected graph G below. The number next to each node is its class label y .

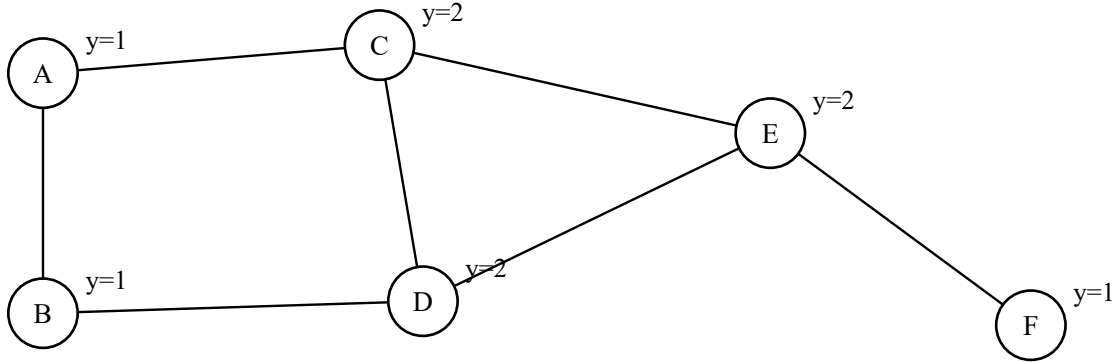


Figure 1. A graph with two classes

Four homophily metrics are defined as follows (C is the number of classes):

$$h_{edge} = |\{(u,v) \text{ in } E : y_u = y_v\}| / |E|$$

$$h_{node} = (1/|V|) \sum_v |\{u \text{ in } N(v) : y_u = y_v\}| / d(v)$$

$$h_{class} = (1/C) \sum_k \max(0, h_k - |V_k|/|V|), \quad h_k = |\{(u,v) \text{ in } E : y_u = y_v = k\}| / |\{(u,v) \text{ in } E : y_u = k\}|$$

$$h_{adj} = (h_{edge} - \sum_k p_k^2) / (1 - \sum_k p_k^2), \quad p_k = \sum_{v \text{ in } V_k} d(v) / 2|E|$$

- Explain why edge homophily can be misleading, and explain what node homophily and adjusted homophily each correct.
- Calculate h_{edge} and h_{node} .
- Calculate h_{class} and h_{adj} , and state whether this graph is homophilic.

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22. (Label Informativeness) Consider the two graphs below. Both have four nodes ($y_A = y_B = 1, y_C = y_D = 2$) and four edges.



Figure 1. Graph G1 (left: A, B, C, D) and Graph G2 (right: E, F, G, H)

Label informativeness measures the mutual information between the labels at the two ends of an edge:

$$LI = I(Y_u ; Y_v) / H(Y_u)$$

$$I(Y_u; Y_v) = \sum_{p,q} P(p,q) \log_2 (P(p,q) / (P(p) P(q))), \quad H(Y_u) = - \sum_p P(p) \log_2 P(p)$$

$P(p,q)$ is the probability that a randomly chosen directed edge goes from a node of class p to a node of class q .

- Explain why edge homophily alone can be a misleading measure of heterophily.
- Calculate h_{edge} and LI for graph G1.
- Calculate h_{edge} and LI for graph G2, and explain which graph a GCN can solve.

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23. (FAGCN) Consider the path graph $A - B - C$ with the node signal $x = [1, 2, 3]^T$ (one feature per node).

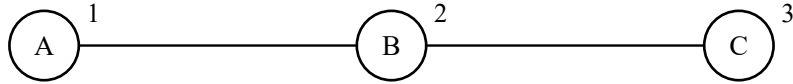


Figure 1. Path graph with signal x

FAGCN defines an enhanced low-pass filter and an enhanced high-pass filter with a scaling parameter $\epsilon \in (0, 1)$:

$$F_L = \epsilon I + D^{-1/2} A D^{-1/2} = (\epsilon + 1) I - L$$

$$F_H = \epsilon I - D^{-1/2} A D^{-1/2} = (\epsilon - 1) I + L$$

Use $\epsilon = 0.5$. The self-gating coefficient of an edge is $\alpha_{ij} = \tanh(g^T [h_i \parallel h_j])$.

- a) Explain what low-frequency and high-frequency signals mean on a graph, and explain why a GCN cannot handle heterophily.
- b) Calculate $D^{-1/2} A D^{-1/2}$, then $F_L \cdot x$ and $F_H \cdot x$.
- c) Given $g^T[h_A \parallel h_B] = 0.4$ and $g^T[h_B \parallel h_C] = -0.8$, calculate the two gating coefficients and explain what a negative coefficient does.

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24. (H2GCN) Consider the undirected graph G below with the embedding $H^{(0)} = [1, 2, 3, 4, 5]^T$ (one value per node, in the order A, B, C, D, E).

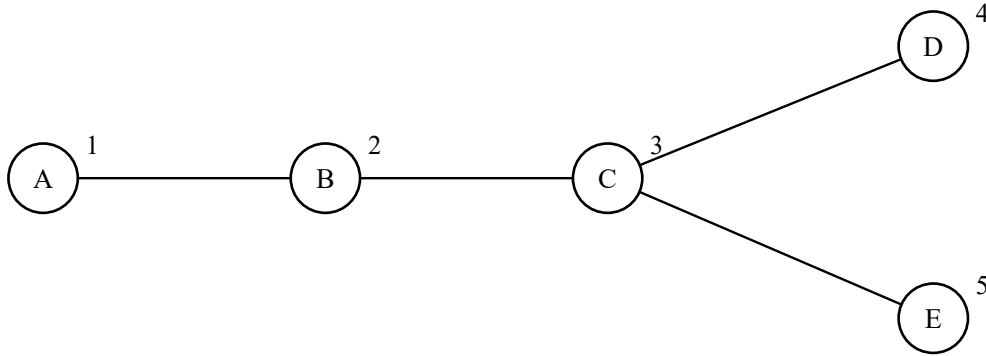


Figure 1. Graph G

H2GCN uses three designs: (D1) ego- and neighbour-embedding separation, (D2) higher-order neighbourhoods, and (D3) a combination of all intermediate representations. With MEAN aggregation and one round $K = 1$:

$$R_1^{(1)} = \text{MEAN}\{ h_u^{(0)} : u \text{ in } N_1(v) \}, \quad R_2^{(1)} = \text{MEAN}\{ h_u^{(0)} : u \text{ in } N_2(v) \}$$

$$H_{final} = \text{CONCAT}(H^{(0)}, R_1^{(1)}, R_2^{(1)})$$

$N_1(v)$ is the set of 1-hop neighbours and $N_2(v)$ is the set of nodes at distance exactly two. Neither set contains v itself.

- a) Explain the three designs D1, D2 and D3, and explain why 2-hop neighbourhoods help under heterophily.
- b) List $N_1(v)$ and $N_2(v)$ for every node and calculate $R_1^{(1)}$ and $R_2^{(1)}$.
- c) Write the final representation H_{final} of every node.

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25. (ACM) Consider the path graph $A - B - C$ with $H^{(0)} = [1, 2, 3]^T$. The Adaptive Channel Mixing (ACM) framework builds three channels per layer.

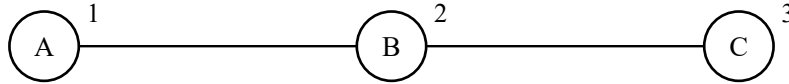


Figure 1. Path graph

With the low-pass filter $H_{LP} = \hat{A}_{rw} = D^{-1}A$, the high-pass filter $H_{HP} = I - \hat{A}_{rw}$, and all weight matrices equal to the identity:

$$H_L = \text{ReLU}(H_L P H), \quad H_H = \text{ReLU}(H_H P H), \quad H_I = \text{ReLU}(I H)$$

$$a_{\sim L} = \text{Sigmoid}(H_L), \quad a_{\sim H} = \text{Sigmoid}(H_H), \quad a_{\sim I} = \text{Sigmoid}(H_I)$$

$$(a_L, a_H, a_I) = \text{Softmax}((a_{\sim L}, a_{\sim H}, a_{\sim I}) / T), \quad T = 1$$

$$H' = \text{ReLU}(\text{diag}(a_L) H_L + \text{diag}(a_H) H_H + \text{diag}(a_I) H_I)$$

- a) Explain the role of each of the three channels, and explain why ACM learns the mixing weights per node instead of once for the whole graph.
- b) Calculate H_L , H_H and H_I .
- c) Calculate the mixing weights of node A and its final representation.

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26. (GGCN) Consider a star graph where node A is connected to B , C and D . The node features are

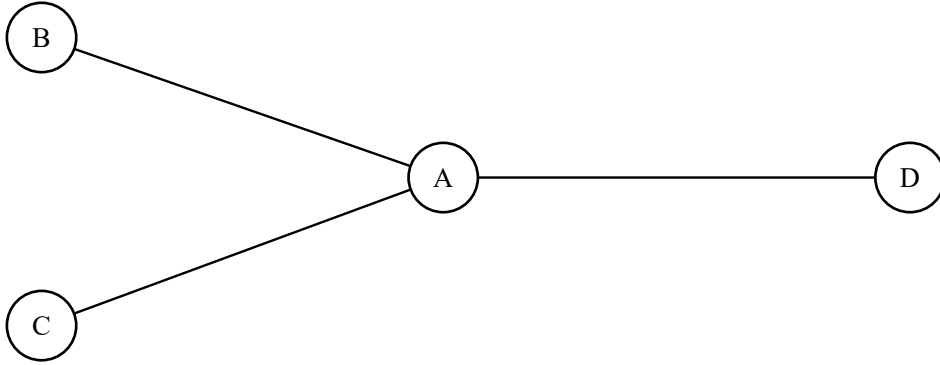


Figure 1. Star graph centred at A

$x_A = [1, 0]^T$, $x_B = [0.8, 0.6]^T$, $x_C = [-1, 0]^T$, $x_D = [0, 1]^T$. All four vectors have unit length.

GGCN replaces the non-negative GCN weight with a signed, feature-based weight:

$$A_{ij}^{new} = \text{Similarity}(x_i, x_j) = \cos(x_i, x_j) \quad (\text{may be negative})$$

$$\text{GCN: } h_A' = \text{MEAN}(x_A, x_B, x_C, x_D)$$

$$\text{GGCN: } h_A' = x_A + \sum_{j \in N(A)} A_{Aj}^{new} x_j$$

- a) Explain why a GCN can only use non-negative edge weights, and explain the claim that heterophily and over-smoothing are two sides of the same coin.
- b) Calculate the three signed edge weights.
- c) Calculate h_A' under GCN and under GGCN, and compare the two results.

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27. (DHGR) Consider the bipartite graph below with node features $x_A = x_B = [1, 0]^T$ and $x_C = x_D = [0, 1]^T$. The labels are $y_A = y_B = 1$ and $y_C = y_D = 2$ (not available to the model).

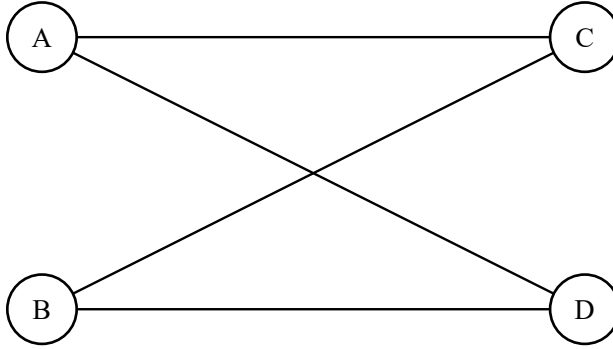


Figure 1. A fully heterophilic graph

DHGR rewires the graph before any GNN is applied. It compares the Neighborhood Feature Distribution of the nodes instead of their raw features:

$$NFD(v_i) = (1/|N(v_i)|) \sum_{v_j \in N(v_i)} x_j, \quad NFD = D^{-1} A X$$

$$S_{ij} = \alpha * \text{sim}(NFD(v_i), NFD(v_j)) + \beta * \text{sim}(NLD(v_i), NLD(v_j))$$

Use $\alpha = 1$, $\beta = 0$ and cosine similarity. Add every non-adjacent pair with $S_{ij} = 1$ and prune every existing edge with $S_{ij} = 0$.

- Explain the difference between model-centric and data-centric approaches to heterophily, and explain why raw features cannot predict edge polarity.
- Calculate h_{edge} of the original graph and the NFD of every node.
- Calculate S_{ij} for all six pairs, perform the rewiring, and calculate the h_{edge} of the rewired graph.

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28. (SignGT) A graph transformer computes attention from node i over three nodes A, B and C . The query, keys and values are

$q_i = [1, 0]$; $k_A = [1, 0]$, $k_B = [0.5, 0]$, $k_C = [-1, 0]$; $v_A = [1, 1]$, $v_B = [2, 0]$, $v_C = [0, 2]$.

Standard self-attention and the signed self-attention (SignSA) of SignGT are

$$\text{Standard: } M_{ij} = \exp(q_i k_j^T) / \sum_k \exp(q_i k_k^T)$$

$$\text{SignSA: } M_{ij}^S = \text{sgn}(q_i k_j^T) * \exp(|q_i k_j^T|) / \sum_k \exp(|q_i k_k^T|)$$

The output is $o_i = \sum_j M_{ij} v_j$ in both cases. Do not scale by $\sqrt{d_k}$.

- a) Explain why a standard graph transformer can only perform smoothing, and explain how SignSA keeps high-frequency information.
- b) Calculate the standard attention weights and the output o_i .
- c) Calculate the SignSA weights and the output o_i , and compare the two results.

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29. (Aggregation and Expressivity) Node colors are encoded as one-hot vectors: yellow = $[1, 0]^T$ and blue = $[0, 1]^T$. Four nodes have the following neighbor multisets:

$N(v_1) = \{ \text{yellow, blue} \}, \quad N(v_2) = \{ \text{yellow, yellow, blue, blue} \}$

$N(v_3) = \{ \text{yellow, blue} \}, \quad N(v_4) = \{ \text{yellow, blue, blue} \}$

Consider three neighborhood aggregation functions applied to the neighbor multiset of a node:

$$\text{Mean}(S) = (1/|S|) \sum_{x \in S} x, \quad \text{Max}(S) = \text{elementwise max}, \quad \text{Sum}(S) = \sum_{x \in S} x$$

- a) Explain what determines the expressive power of a message passing GNN, and why.
- b) Calculate Mean, Max and Sum for all four nodes.
- c) State which pair mean pooling fails to distinguish, which pair max pooling fails to distinguish, and whether sum pooling distinguishes both pairs.

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30. (GIN) Consider an undirected graph G with an adjacency matrix A and an initial node feature matrix $H^{(0)}$ of one-hot colors as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

A Graph Isomorphism Network (GIN) layer is defined as:

$$h_v^{(k)} = \text{MLP}\left((1 + \varepsilon) \cdot h_v^{(k-1)} + \sum_{u \in N(v)} h_u^{(k-1)}\right)$$

where $\varepsilon = 0.1$. Assume the MLP is a single linear layer followed by ReLU, $\text{MLP}(x) = \sigma(xW)$, with

$$W = \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}$$

and σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

- a) Explain why GIN is the most expressive message passing GNN.
- b) Calculate the aggregated values inside the MLP for all nodes at layer $k = 1$.
- c) Calculate $H^{(1)}$, and compare the aggregation with that of a standard GCN layer.

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31. (1-WL Test) Consider two undirected graphs G_1 and G_2 with five nodes each, given in the following figures.

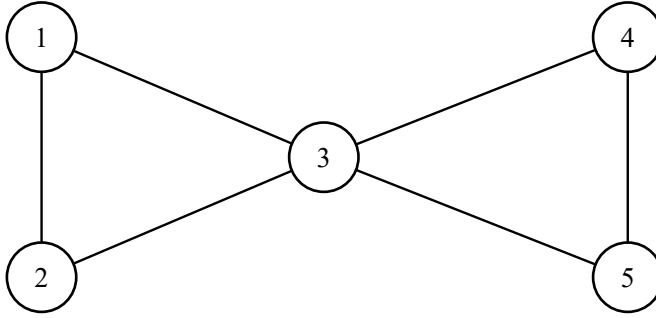


Figure 1. Graph G1

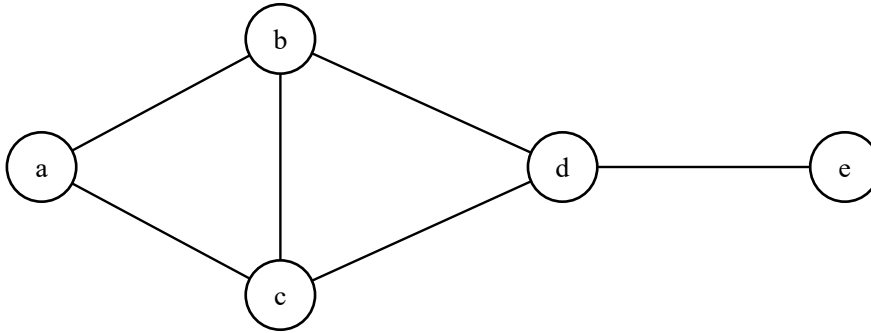


Figure 2. Graph G2

The 1-WL (Weisfeiler-Lehman) color refinement algorithm assigns every node the same initial color $c^{(0)}(v) = 1$, and then repeats:

$$c^{(k+1)}(v) = \text{HASH}(\text{left}(c^{(k)}(v), \{c^{(k)}(u) \}_{u \in N(v)} \text{right}))$$

where HASH gives a new color to each distinct pair, and the neighbor colors form a multiset.

- a) Explain the 1-WL color refinement algorithm and what its result can and cannot prove.
- b) Run two rounds of color refinement on G_1 and report the color classes.
- c) Run the same on G_2 and decide whether G_1 and G_2 can be isomorphic.